AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: GEN26-A0H-I223

Team Name: Syntax Terror

Domain: Intelligent Systems and Digital Futures

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Use Cases
9. Impacts and Benefits
10. Sustainable Development Goals
11. Individual Member Contribution

Problem Statement

Voice notes are effortless to send, but completely impossible to search, organize, or retrieve.

Problem Understanding

- **Who experiences this problem?**
  College students, freelancers, and small business owners who rely on instant messaging and handle large volumes of audio messages daily.

- **What causes the problem?**
  Messaging apps only index typed text. Audio files are not indexed, making spoken words entirely unsearchable

- **How does it affect users or stakeholders?**
  The messaging platforms themselves like WhatsApp or Meta because this tool works well within their export features. It strengthens the ecosystem around these platforms.

Existing Gaps & Challenges

- **Existing gap or limitation 1 :**
  Running speech to text locally without draining phone batteries or causing device lag.

- **Existing gap or limitation 2 :**
  Accurately parsing real world audio with background noise, accents and mixed languages.

- **Main challenge that remains unsolved :**
  Running accurate speech to text models locally on mobile hardware without draining battery or violating the encrypted app constraint.

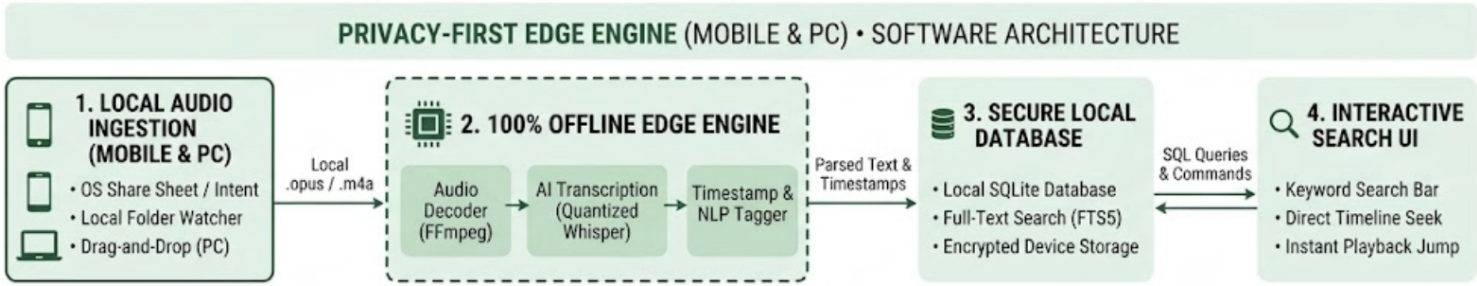
Our Proposed Solution

- **Core idea or product :**
  We planned to built a companion app for mobile and PC that adds a search bar to your WhatsApp voice notes.

- **How the solution works at a high level :**
  The offline-only companion app takes local . opus audio files that you share through the OS share sheet then uses on device Edge AI to transcribe the audio into text and finally stores that text with exact timestamps in a local SQLite database.

- **Main value delivered to users :**
  The offline only companion app removes the need, for audio scrubbing. Users can search a keyword find the voice note and jump directly to the second when the word was spoken while keeping data privacy safe.

Technical Approach



The workflow runs completely offline to keep your data private. Voice notes are brought in using the OS share menu and sent to the Edge Engine. The Edge Engine turns the audio into text by using a version of the Whisper model. It also adds timestamps to the text so everything is properly organized. This data gets saved in a SQLite database. The user interface can then quickly search for words and jump right to the part of the audio that matches—no cloud servers needed. Everything stays on your device.

Tech Stack

Frontend: Flutter, Tauri

Database: SQLite with FTS5

Backend: Local device processing

AI/ML, APIs, or Hardware: Whisper.cpp, ONNX

Innovation & Uniqueness

- **What is new or different?**
  What is new is a decentralized on device indexing pipeline. By coupling quantized speechto text Whisper.cpp with an embedded SQLite Full-Text Search engine the system automatically indexes spoken words .

- **Why is this approach suited to the problem?**
  Users use messaging apps to keep their messages private. Sending voice messages to cloud servers on the internet breaks that promise and can cause problems with rules about data. Doing the work, on the phone or computer keeps all information completely private and doesn't need the internet at all.

- **What is the key differentiator?**
  Global cross-chat search, with exact timeline jumping. Our engine does more than just transcribe. It acts as an index. It searches through all chat histories. Users can jump straight to the second a matching keyword was spoken. No need to scroll or guess. Just find the moment. Fast. Accurate. Right where it happened.

Feasibility & Scalability

- **Feasibility:** The app is built with Flutter so that the app can run on devices. The app uses FFmpeg to decode right on the device. The app also uses a quantized version of Whisper.cpp to perform transcription with Edge AI, all offline. The app takes the text that the app has produced along with millisecond timestamps. Stores them, in a local SQLite database.

- **Scalability:** Because all processing happens on the users device we don't need any servers at all. That means we can support a number of users without having to build more infrastructure. The system is flexible. Can easily be updated to index audio from Telegram, Signal or even voice memos saved directly on the device.

- **Risks:** The main concern is that battery life and memory can be quickly drained on entry level devices. To handle this we use lightweight AI models that are quantized to INT4 taking up less than 150MB of RAM. We also make sure that heavy background transcription tasks only run when the device is not, in use or is connected to a charger. This helps keep performance stable and saves power.

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1:** A lawyer receives sensitive case details from clients by way of WhatsApp voice notes. The lawyer must search for facts many months after the call. However strict data privacy laws forbid the lawyer from uploading client audio to third party cloud transcription services. The edge engine gives the lawyer instant keyword retrieval yet keeps all information strictly on the lawyer's device.

- **Use case 2:** I am a student working on a software project with teammates who keep sending rambling voice notes that explain code logic or presentation ideas. Of listening to hours of audio by hand during a late night study session I can type a keyword such, as database and jump straight to the exact second where database is mentioned.

- **Use case 3:** A field reporter does interviews. Receives audio drops in remote places that have no internet. The field reporter must quickly. Look through these audio drops to check quotes before writing the story. The 100 percent offline system keeps the field reporter working without WiFi. It keeps the sources safe, from cloud leaks.

Impacts and Benefits

- **Benefit to the target users :**
  Users get back hours of lost productivity by searching and jumping straight to exact keywords tucked inside long voice notes. Beyond saving time Users feel complete peace of mind because their highly sensitive personal and professional conversations are handled entirely on-device which fully removes the danger of cloud data leaks.

- **Social, economic, or environmental impact :**
  This architecture drives both progress and environmental progress by giving everyone free access, to top AI tools without expensive cloud subscription paywalls.

- **Expected measurable outcome :**
  The system makes sure that no audio or text is ever sent to servers. It really makes finding information faster going from several minutes of listening to, under two seconds. It also lets the system grow much as possible without costing anything extra for each new person who uses it.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [9]	Target [9.4]	[Industry, Innovation and Infrastructure]	[Promotes sustainable edge computing]
SDG [8]	Target [8.2]	[Decent Work and Economic Growth]	[Boosts daily professional productivity]

Individual Member Contribution

Member	Contribution or responsibility
Lakshieth K K	Team Leader , Concept Thinking
Guhannathan B L	Design , Formatting
Dharun B	Chart Work, Flowcharts , Visual Drawings
Tejesh Satya	Pitch Strategy , Brainstorming

Yukti ID

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